

Bayesian and Advanced Modeling of Bike-Sharing Demand

SDS 2 Endterm Project Report

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Dataset: UCI Bike Sharing Dataset

1. Introduction

Bike-sharing systems are an important part of urban mobility because they provide flexible short-distance transportation. Their operational challenge is demand uncertainty: the number of bike rentals changes by hour, day, season, and weather. This project analyzes real-world hourly bike-sharing demand and follows a complete data science workflow.

2. Problem Statement

The main research question is: how do time, weather, seasonality, and working-day structure influence hourly bike-sharing demand, and how accurately can demand be forecasted?

The project aims to describe demand patterns, estimate uncertainty using Bayesian inference, model temporal behavior using time series analysis, study short-term state transitions using a Markov chain, and build a machine learning workflow for prediction.

3. Dataset Description

The selected dataset is the Bike Sharing Dataset from the UCI Machine Learning Repository. The hourly file contains bike rental records from the Capital Bikeshare system in Washington, D.C. for 2011-2012. It includes calendar variables, weather variables, casual and registered user counts, and the total rental count.

Metric	Value
Rows	17,379
Columns after preprocessing	21
Date range	2011-01-01 to 2012-12-31
Target variable	cnt: total hourly rentals
Mean hourly rentals	189.46
Median hourly rentals	142
Minimum / maximum	1 / 977

The dataset satisfies the project requirements because it is a real-world dataset, has more than 1,000 observations, includes multiple variables, and has clear temporal structure.

4. Data Cleaning and Preprocessing

The notebook loads either an uploaded dataset or the default UCI dataset. For the submitted analysis, the UCI hourly bike-sharing file was used. The preprocessing pipeline parses the date, creates a unified datetime field, selects the target variable, creates calendar features, checks missing values, checks duplicate rows, and prepares daily totals for time series modeling.

Check	Result
Missing values	0 in the core columns used for analysis
Duplicate rows	0
Target cleaning	cnt converted to numeric and clipped at 0
Feature engineering	datetime, log_cnt, is_peak_hour, date
Time-series aggregation	hourly counts aggregated to daily totals

5. Exploratory Data Analysis

The EDA shows that bike-sharing demand is highly structured. Demand has a right-skewed distribution, strong hourly patterns, seasonal differences, and weather-related variation. Peak hours align with commuting behavior.

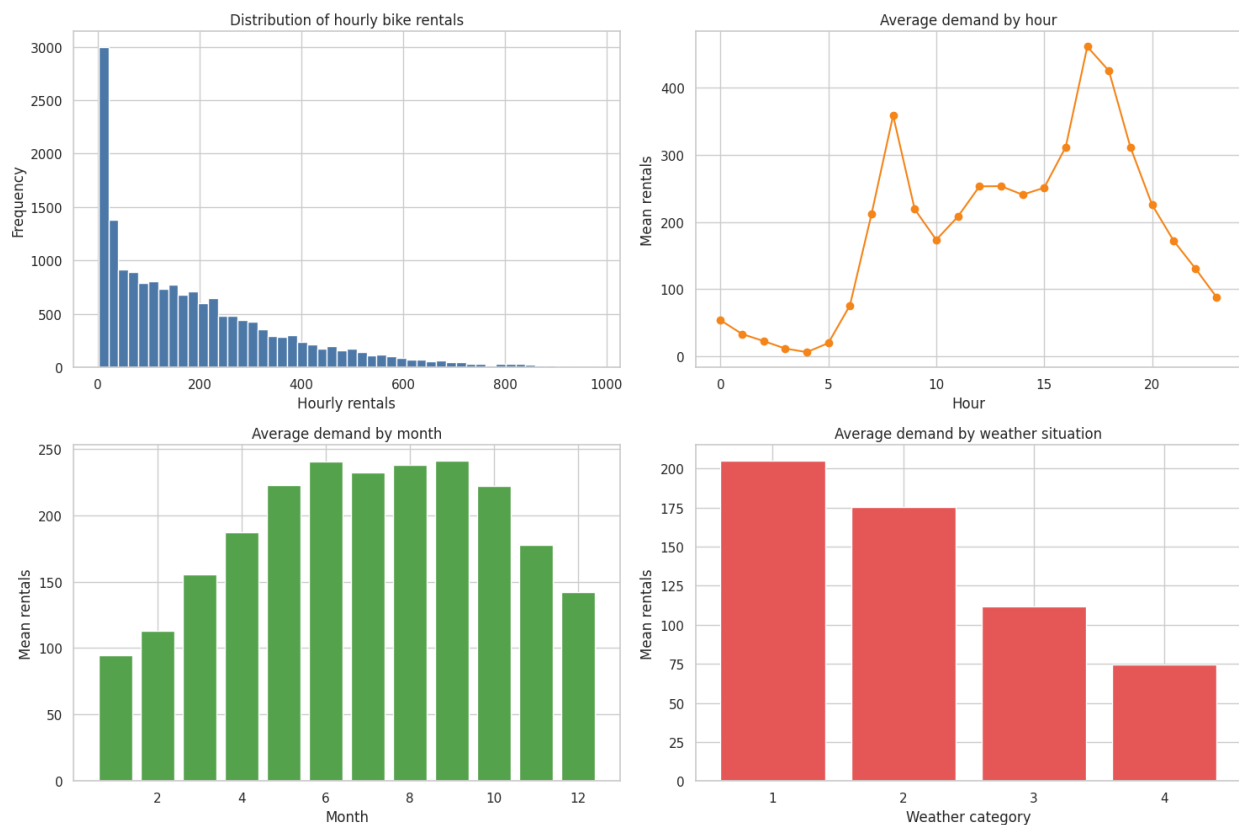


Figure 1. Distribution, hourly pattern, monthly pattern, and weather effect.

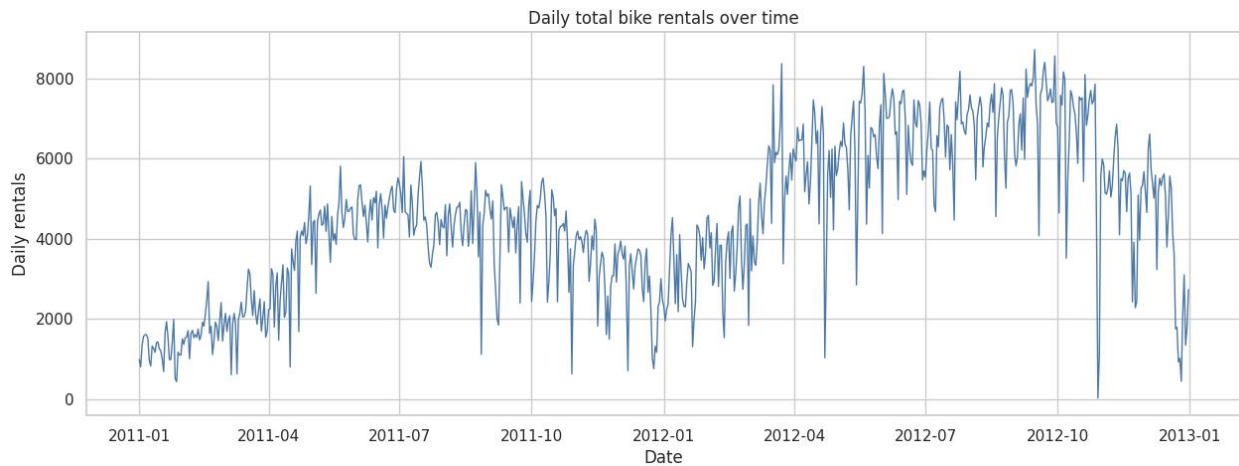


Figure 2. Daily total bike rentals over time.

The highest average demand occurs during commuter hours. Weather category 1 has the highest demand, while worse weather conditions reduce rentals. These patterns justify using temporal and probabilistic methods.

6. Bayesian Analysis

Bayesian inference is applied using a Gamma-Poisson model. Hourly rentals are non-negative count data, so a Poisson likelihood is a natural first model. A Gamma prior is used for the positive demand rate because it is conjugate to the Poisson likelihood.

Model:

$\text{cnt}_i | \lambda_{\text{g}} \sim \text{Poisson}(\lambda_{\text{g}})$

$\lambda_{\text{g}} \sim \text{Gamma}(\alpha_0, \beta_0)$, with $\alpha_0 = 2$ and $\beta_0 = 0.01$

Group	n	Observed mean	Posterior mean	95% credible interval
Non-working day	5,514	181.41	181.41	[181.05, 181.76]
Working day	11,865	193.21	193.21	[192.96, 193.46]

Weather	n	Posterior mean	95% credible interval
1	11,413	204.87	[204.61, 205.13]
2	4,544	175.17	[174.78, 175.55]
3	1,419	111.58	[111.03, 112.13]
4	3	74.75	[65.30, 84.83]

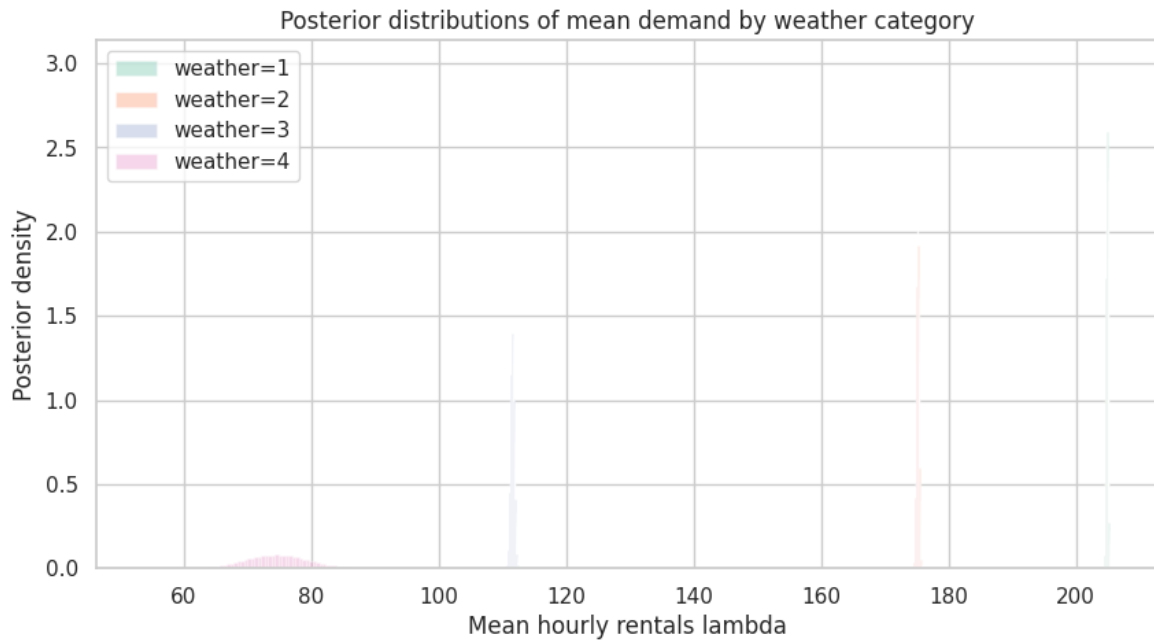


Figure 3. Posterior distributions of mean demand by weather category.

The posterior probability that working-day demand is higher than non-working-day demand is 1.0000 in the notebook output. This supports the interpretation that commuting activity increases expected rentals.

7. Advanced Modeling

7.1 Time Series Analysis

The hourly observations were aggregated into daily totals and modeled with SARIMAX. This method is appropriate because demand is chronological and shows repeated temporal patterns. The final test metrics were MAE = 1515.60 and RMSE = 2092.53 for daily rentals.

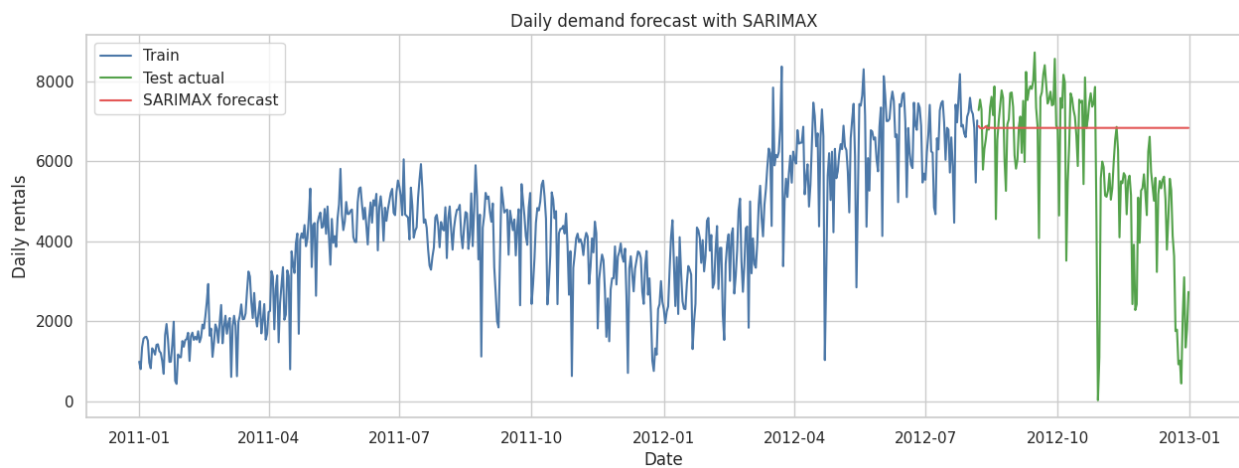


Figure 4. SARIMAX daily demand forecast.

7.2 Markov Chain Modeling

Demand was divided into low, medium, and high states. The Markov transition matrix estimates the probability of moving from one state in hour t to another state in hour $t+1$.

Current state	Next low	Next medium	Next high
Low	0.665	0.254	0.081
Medium	0.242	0.491	0.267
High	0.093	0.259	0.648

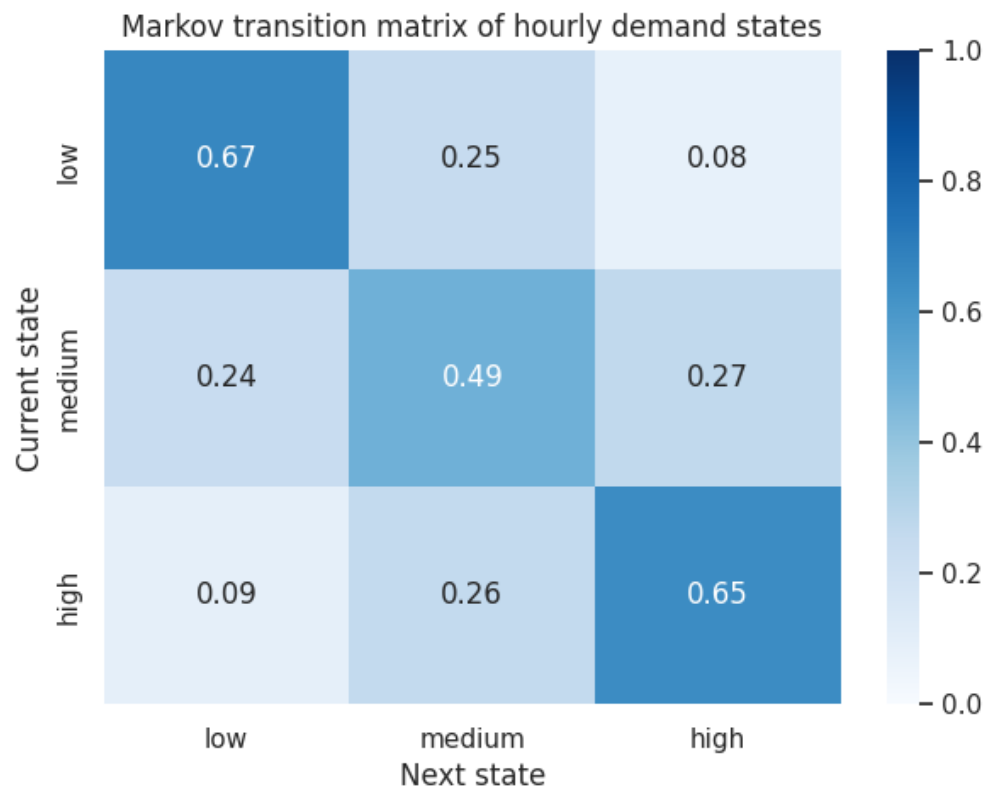


Figure 5. Markov transition matrix heatmap.

The diagonal probabilities are high for low and high states, meaning demand tends to persist from one hour to the next. This is useful for operational planning because peak demand usually lasts more than one hour.

7.3 Machine Learning Workflow

The machine learning workflow predicts hourly demand using calendar and weather features. A time-ordered train/test split is used to avoid leakage from future observations into the training set.

Model	MAE	RMSE	R-squared
HistGradientBoosting	49.58	71.49	0.895
RandomForest	57.06	84.33	0.854

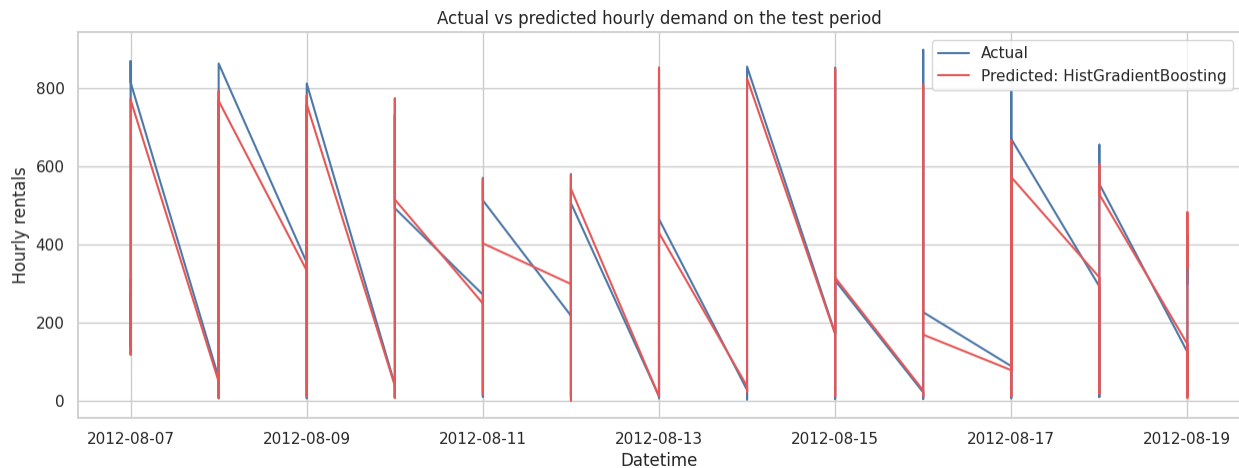


Figure 6. Actual vs predicted hourly demand on the test period.

HistGradientBoosting achieved the lowest RMSE and highest R-squared, so it is the best-performing model in this comparison. The result shows that calendar and weather variables contain useful predictive information.

8. Results and Interpretation

The project finds that bike-sharing demand is mainly driven by time-of-day structure, weather conditions, and seasonal patterns. The Bayesian model supports a credible difference between working-day and non-working-day mean demand. The time series model captures daily temporal behavior, while the Markov chain summarizes short-term persistence between demand states. The machine learning workflow provides a strong predictive benchmark.

9. Limitations

The dataset covers one city and two years, so the results should not automatically be generalized to every bike-sharing system. Weather variables are normalized and do not capture all local conditions. The Bayesian model is intentionally simple and focuses on group-level demand rates; a stronger extension would be a full Bayesian regression model with multiple predictors.

10. Conclusion

This project demonstrates a complete data science workflow using a real-world dataset. It includes data acquisition, preprocessing, EDA, Bayesian inference, time series analysis, Markov chain modeling, machine learning prediction, visualization, and interpretation. The selected methods match the count-based and temporal structure of the dataset and satisfy the SDS2 project criteria.

11. References

Fanaee-T, H. (2013). Bike Sharing Dataset. UCI Machine Learning Repository.
<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

Gelman, A. et al. Bayesian Data Analysis.

Box, G. E. P. et al. Time Series Analysis: Forecasting and Control.